

HealthConnect Analytics Testing & Refinement Report

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Week 7 – Data Analytics Track

1. Introduction

Week 7 focused on testing, refining and validating the Data Analytics outputs developed during Week 6. The objective was to confirm the accuracy of key KPIs and analytical findings, validate dashboard outputs against the underlying dataset, identify analytical limitations, improve dashboard interpretation and prepare the analytics component for the next stage of the HealthConnect solution.

The Week 7 process followed the sequence of **testing, identifying issues, refining, retesting, validating and documenting results**.

2. Week 6 to Week 7 Transition

The main Week 6 output was an advanced Power BI dashboard analysing appointment no-show behaviour across:

- Previous no-show history
- Reminder status
- Appointment type
- Age group
- Booking lead time
- Distance to clinic
- Waiting time

The main finding requiring further validation was the observed relationship between **previous no-show history and current appointment no-show behaviour**.

Week 7 therefore focused on validating the key KPIs and analytical findings against the underlying HealthConnect appointment dataset, testing dashboard outputs and refining interpretations where necessary.

3. Testing Approach

The following analytical components were tested:

1. Overall appointment KPIs
2. No-show rate calculation
3. Previous no-show history
4. Reminder status
5. Appointment type
6. Age group
7. Booking lead time
8. Distance and waiting time
9. Dashboard filters
10. Dashboard visual consistency and interpretation

Testing focused on comparing expected results with actual results, identifying issues or limitations, applying appropriate refinements and confirming results through retesting. This follows the testing-record approach required for Week 7.

4. KPI Validation

The main dashboard KPIs were checked against the underlying HealthConnect appointment records.

KPI	Validated Result
Total Appointments	5,000
No-Show Count	2,423
No-Show Rate	48.46%
Attendance Rate	46.3%
Cancellation Rate	5.3%

The KPI values were consistent with the underlying dataset and were retained in the refined dashboard.

Validation Result

Status: Pass

No major calculation error was identified in the overall KPI measures.

5. Validation of Important Analytical Findings

5.1 Previous No-Show History

The Week 6 finding that previous no-show history is associated with current no-show behaviour was tested using a validation table in Power BI.

Previous No-Shows	Total Appointments	No-Show Count	No-Show Rate
0	2,921	1,271	43.51%
1	1,548	828	53.49%
2	438	260	59.36%
3	78	53	67.95%
4	12	8	66.67%
5	3	3	100.00%
Total	5,000	2,423	48.46%

The validation confirmed the pattern shown in the Week 6 dashboard.

The no-show rate generally increased as previous no-show history increased from 0 to 3 previous no-shows.

However, the groups with four and five previous no-shows contain very few appointments. Therefore, their percentages were treated cautiously rather than being interpreted as stable estimates.

Testing Result: Pass, with a small-sample limitation.

5.2 Reminder Status

The Week 6 dashboard showed:

- **No reminder:** 51.39% no-show rate
- **Reminder sent:** 47.36% no-show rate

This finding was retained after validation.

However, the relationship is interpreted as an **observed association rather than a causal effect**, because the dataset is observational.

Testing Result: Pass, with an observational-data limitation.

5.3 Appointment Type

The analysis showed differences in no-show rates across appointment types.

The observed rates were:

- Follow-up: 51.23%
- Diagnostic Test: 49.75%
- Specialist Consultation: 47.44%
- General Consultation: 46.54%

The finding was retained as a useful area for continued monitoring.

Testing Result: Pass.

5.4 Age Group

Differences in no-show rates were observed across age groups.

The results were retained as descriptive findings that can support patient-segment monitoring.

These differences were not interpreted as evidence that age directly causes missed appointments.

Testing Result: Pass.

5.5 Booking Lead Time

The Week 6 analysis showed a broad increase in observed no-show rates at longer booking lead times.

The finding was retained as a potential indicator for appointment-management monitoring.

Further analysis would be required before treating booking lead time as a direct cause of missed appointments.

Testing Result: Pass.

5.6 Distance and Waiting Time

Distance to the clinic and waiting time were reviewed as operational factors.

The dashboard continued to present average distance and waiting time by appointment outcome.

These variables were retained for monitoring, while avoiding strong causal conclusions from average-level comparisons.

Testing Result: Pass, with an interpretation limitation.

6. Dashboard Testing and Refinement

The Week 6 dashboard was duplicated to create the Week 7 testing and refinement version. This allowed the original Week 6 dashboard to remain available as a baseline for comparison.

The Week 7 dashboard was refined rather than completely recreated.

Main refinements included:

- Addition of a validation table for previous no-show history.
- Refinement of the previous no-show chart title to “**No-Show Rate by Previous No-Show History.**”
- Addition of a caution note stating that small groups with 4–5 previous no-shows should be interpreted carefully.
- Retention of validated KPI cards.
- Retention and refinement of relevant analytical visuals.
- Continued use of slicers for appointment type, appointment outcome, gender and age group.
- Refinement of recommendations to reflect validated findings and analytical limitations.

The final Week 7 dashboard provides both analytical findings and supporting validation evidence.

7. Testing and Refinement Summary

Component Tested	Expected Result	Actual Result	Status	Action
KPI calculations	Dashboard KPIs match underlying data	Values matched	Pass	Retained
Previous no-show history	Dashboard pattern reproduced	Pattern reproduced	Pass	Added small-sample caution
Reminder status	Rates remain consistent	Rates consistent	Pass	Avoid causal interpretation
Appointment type	Rates remain consistent	Differences remained	Pass	Retained
Age group	Rates remain consistent	Differences remained	Pass	Retained
Booking lead time	Pattern remains visible	Broad upward pattern remained	Pass	Retained for monitoring
Distance/waiting time	Operational pattern remains interpretable	Descriptive comparison remained	Pass	Interpret cautiously
Dashboard filters	Filters should allow segmentation	Filters available and functional	Pass	Retained

The testing process resulted in refinement rather than major restructuring because the key analytical outputs remained consistent with the underlying data.

8. Validated Key Findings

The Week 7 testing process supported the following findings:

1. Previous no-show history

Current no-show rates generally increase as previous no-show history increases. This makes previous no-show history an important indicator for appointment-management monitoring.

2. Reminder status

Appointments with a recorded reminder had a lower observed no-show rate than appointments without a recorded reminder.

This is an association and should not be interpreted as proof that reminders caused the difference.

3. Appointment type

No-show rates vary across appointment types, with Follow-up appointments showing the highest observed rate among the categories analysed.

4. Age group

No-show rates vary across age groups and can therefore support continued segmentation and monitoring.

5. Booking lead time

Longer booking lead times show a broad increase in observed no-show rates.

6. Operational factors

Distance and waiting time remain useful operational variables to monitor when considering the overall patient appointment experience.

9. Updated Recommendations

Based on the validated findings, the following actions are recommended:

1. **Prioritise patients with previous no-show history** for additional appointment confirmation and engagement.
 2. **Strengthen reminder processes** and continue monitoring reminder coverage and observed appointment outcomes.
 3. **Monitor higher-risk appointment segments**, particularly appointment types and patient segments with relatively higher observed no-show rates.
 4. **Monitor longer booking lead times** and consider additional confirmation closer to the appointment date.
 5. **Continue monitoring operational factors**, including distance to the clinic and waiting time.
 6. **Use small segments cautiously** and avoid making strong decisions from groups with very few observations.
 7. Continue validating analytical findings before they are used for automated or high-impact decision-making.
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10. Cross-Track Testing Status

A cross-track validation activity was identified as a Week 7 requirement, particularly in relation to sharing validated analytical findings that could inform other HealthConnect workstreams. The Data Analytics findings were prepared for cross-track sharing, but feedback was not received within the Week 7 submission period.

Therefore, **no cross-track testing outcome or refinement is claimed in this report.**

The validated findings remain available for follow-up with the relevant HealthConnect track in the next stage of the project.

This is documented as an outstanding dependency rather than presenting an uncompleted collaboration as completed. The Week 7 requirements specifically call for meaningful cross-track testing evidence, including the dependency, testing activity, finding, action and retest result.

11. Limitations

The following limitations remain:

- The dataset is observational, so associations should not be interpreted as causal relationships.
- The groups with four and five previous no-shows contain very few appointments.
- Distance to clinic and waiting-time variables contain limited missing values.
- Average-level comparisons may not capture all factors influencing appointment attendance.
- The analysis identifies patterns and potential risk indicators but does not independently establish the causes of missed appointments.
- Cross-track validation remains outstanding because feedback was not received before submission.

12. Evidence Produced

The Week 7 Data Analytics work produced:

- Refined Power BI dashboard.
- KPI validation.
- Previous no-show validation table.
- Analytical testing and refinement evidence.
- Updated findings and recommendations.
- Documented analytical limitations.

- Week 6 dashboard retained as the baseline.
- Week 7 dashboard created as the refined/testing version.

The Week 7 requirements call for a track-specific testing/refinement report or notebook together with testing evidence, refined dashboard/visualisations, validated findings, recommendations, limitations and supporting evidence.

13. Week 8 Readiness

Before progressing to Week 8, the following items should remain available:

- Final Week 7 Power BI dashboard.
 - Week 6 baseline dashboard.
 - Validation table and testing evidence.
 - Final Week 7 report.
 - Documented limitations and outstanding dependencies.
 - Cross-track validation for follow-up when the relevant team provides feedback.
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14. Conclusion

Week 7 strengthened the HealthConnect Data Analytics work by moving from advanced analysis into structured testing, validation and refinement.

The main KPIs were validated against the underlying appointment dataset, important analytical findings were tested, dashboard outputs were refined, and limitations were documented.

The strongest validated finding was the association between **previous no-show history and current no-show behaviour**, with no-show rates generally increasing as previous no-show history increased. Other findings relating to reminder status, appointment type, age group and booking lead time were also retained as useful areas for continued monitoring.

The final Week 7 dashboard provides a clearer and more validated analytical view of appointment attendance while recognising the limitations of the available data.

Overall, the Week 7 Data Analytics component provides a tested and refined foundation for the next stage of the HealthConnect project.